Exp No:1.a) Analyze the trend of data science job postings over the last decade.

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

data = {'Year': list(range(2010, 2021)), 'Job Postings': [150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400, 4200]}

 df = pd.DataFrame(data)

 plt.plot(df['Year'], df['Job Postings'], marker='o')

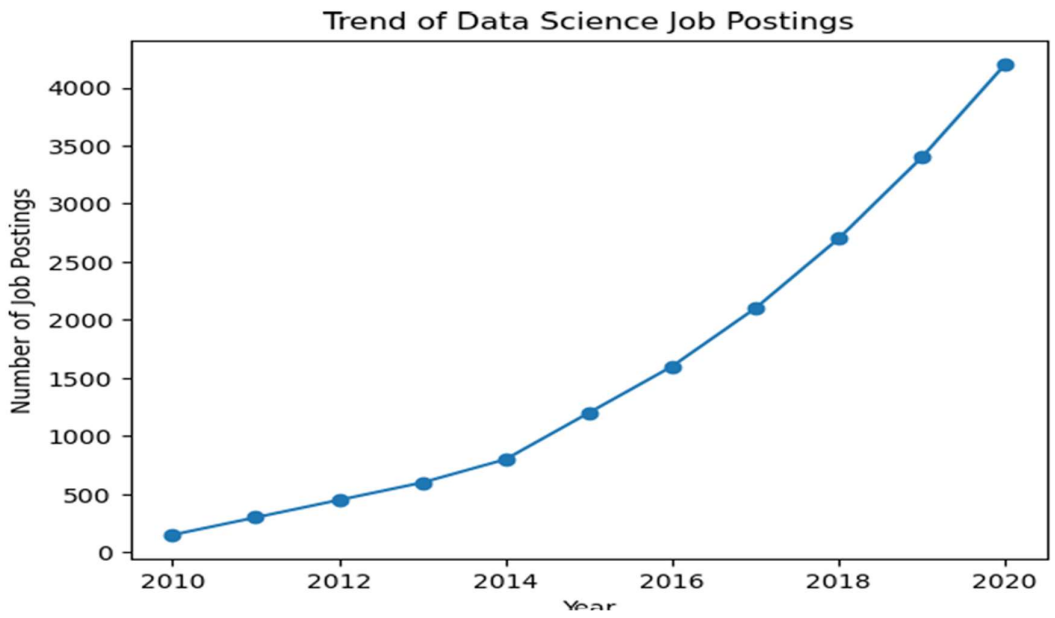
 plt.title('Trend of Data Science Job Postings')

plt.xlabel('Year')

plt.ylabel('Number of Job Postings')

plt.show()
```

Output:



Result:

The trend of data science job postings over the last decade was analysed.

Exp No:1.b) Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

Code:

```
roles = ['Data Analyst', 'Data Engineer', 'Data Scientist', 'ML Engineer', 'Business Analyst']

counts = [300, 500, 450, 200, 150]

plt.bar(roles, counts)

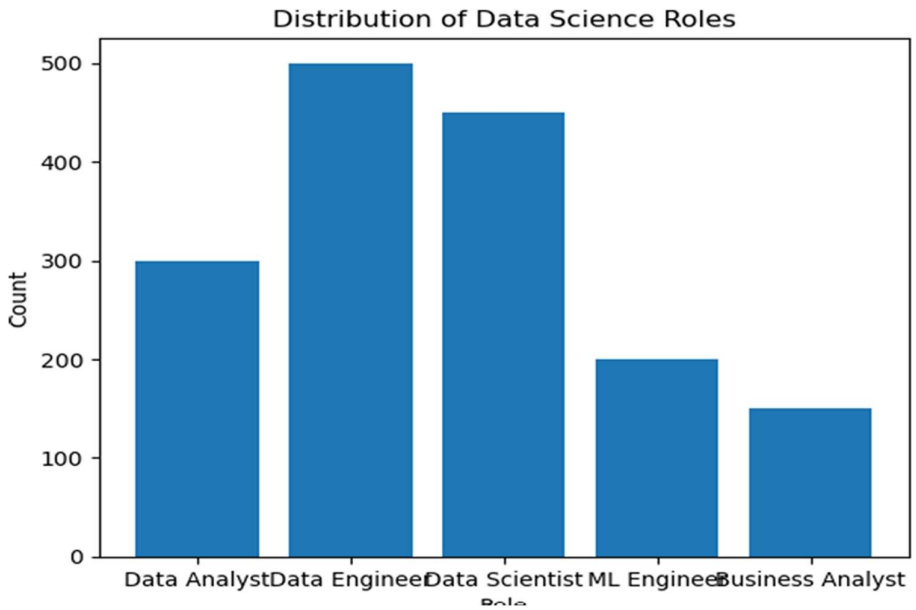
plt.title('Distribution of Data Science Roles')

plt.xlabel('Role')

plt.ylabel('Count')

 plt.show()
```

Output:



Result:

the distribution of various data science roles were analyse and this code was executed successfully.

Exp No:1.c) Conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.

Code:

structured_data = pd.DataFrame({ 'ID': [1, 2, 3], 'Name': ['Alice', 'Bob', 'Charlie'], 'Age': [25, 30, 35] })

print("Structured Data:\n", structured_data)

unstructured_data = "This is an example of unstructured data. It can be a piece of text, an image, or a video file." print("\nUnstructured Data:\n", unstructured_data)

semi_structured_data = {'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}

print("\nSemi-structured Data:\n", semi_structured_data)

Output:

```
Structured Data:
    ID     Name  Age
0   1    Alice   25
1   2      Bob   30
2   3  Charlie   35

Unstructured Data:
 This is an example of unstructured data. It can be a piece of text, an
image, or a video file.

Semi-structured Data:
 {'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}
}
```

Result:

an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given were analysed.

Exp No:1.d) Conduct an experiment to encrypt and decrypt given sensitive data**.**

Code:

```
from cryptography.fernet import Fernet

key = Fernet.generate_key()

f = Fernet(key) token = f.encrypt(b"Rajalakshmi Engineering College")

 token

 b'...'

 f.decrypt(token)

b'Rajalakshmi Engineering College'

 key = Fernet.generate_key()

cipher_suite = Fernet(key)

plain_text = b"Rajalakshmi Engineering College."

cipher_text = cipher_suite.encrypt(plain_text )

decrypted_text = cipher_suite.decrypt(cipher_text)

print("Original Data:", plain_text)

print("Encrypted Data:", cipher_text)

print("Decrypted Data:", decrypted_text)
```

Output:

```
Original Data: b'Rajalakshmi Engineering College.'
Encrypted Data: b'gAAAAABmmnHEhy1ZC1-Dox-URNAvYqFFGfJemMb_fKmXqhATe3AAn
XEOnSzaNRINI4zAbNK8lgx2pprrNn-j6agbWDf30OwbrZsBuz8fnDdX7N35WwloDu3m6pVv
kwv8GYDSI8Fov1tV'
Decrypted Data: b'Rajalakshmi Engineering College.'
```

Result:

Thus **,** experiment to encrypt and decrypt given sensitive data was analysed and this code executed successfully.